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A comprehensive assessment of tandem repeat genotyping methods for Nanopore long-read genomes

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Abstract

Background

Tandem repeats play critical roles in human disease and phenotypic diversity but are among the most challenging classes of genomic variation to measure accurately. Long-read sequencing has the potential to accurately characterize long and complex tandem repeats. While an increasing number of genotyping methods are available, no systematic effort has been undertaken to evaluate their usability, accuracy, and performance across motifs and allele lengths.

Results

We reviewed 25 bioinformatic tools and selected seven actively maintained methods for benchmarking using publicly available Oxford Nanopore genome sequencing data from more than 100 individuals. We assessed performance across 43,009 genome-wide tandem repeat loci using four complementary strategies: concordance with haplotype-resolved Human Pangenome Reference Consortium assemblies, Mendelian consistency, cross-tool consistency, and sensitivity to molecularly confirmed pathogenic expansions. Most methods achieved high concordance with assemblies, with higher accuracy using R10 Oxford Nanopore pore chemistry than older R9 chemistry. Accuracy declined with increasing allele length, and most tools performed worse on homopolymers, heterozygous loci, and alleles differing from the reference genome. Assembly concordance and Mendelian consistency did not predict sensitivity to pathogenic expansions, suggesting that these metrics captured distinct aspects of performance.

Conclusions

No single genotyper performs consistently best across all assessments, but strong contenders emerge in each. Our results demonstrate that length accuracy overestimates tandem repeat genotyping performance. Sequence-level benchmarking is essential for selecting tools best-suited for population studies and clinical diagnostics. This work provides practical guidance for tool selection and highlights key priorities for future long-read tandem repeat genotyping method development.

Background

While tandem repeats (TRs) are among the most variable and mutable regions in the genome, with mutation rates on the order of 10^{-5} per generation [1–4], over three orders of magnitude higher than single-nucleotide variants, they are among the most challenging classes of variation to measure accurately. TRs include short tandem repeats (STRs, 1-6 bp motifs) and variable number tandem repeats (VNTRs, 7+ bp motifs). Today, more than 70 TR loci are associated with monogenic diseases in humans [5–7], and many more are associated with increased risk of complex disease or other phenotypic variation [8–12]. TRs remain understudied due to

challenges in accurately genotyping them at scale.

Genome-wide studies using short-read sequencing (SRS) approaches have improved our understanding of the scale of TR variation and allowed numerous new disease discoveries. But, while bioinformatic method development for TR genotyping using SRS has been highly active [13], these methods are limited to evaluating short, relatively simple loci due to their inability to span longer repeat alleles and fully characterize sequence variation within these alleles.

Long-read sequencing (LRS) has dramatically increased our ability to fully characterize long and complex repeat structures by revealing previously unknown features of TR biology, including motif diversity and novel mutation patterns within TR loci. This is possible because the longer read lengths generated by LRS more easily span longer repeat structures, align better within repetitive regions, and are more easily assembled than SRS data. These data can then be more easily used to resolve complex TR arrangements, assess somatic variation, phase alleles into haplotypes, and identify base modifications using a single data source. Still, challenges remain in genotyping TRs using LRS data, including higher per-base error rates, increased homopolymer errors, and, historically, sequencing cost.

Accurate determination of allele sequence, rather than just repeat number and allele size, is important for diagnosing and predicting the prognosis of many repeat expansion disorders. For example, within *RFC1*, specific motifs are considered benign at any size, while other motifs are pathogenic when expanded [14]. In the *FMR1* gene, AGG interruptions of the CGG repeat are stabilizing [14,15]. These interruptions reduce the probability that a parent with a premutation allele close to the threshold for pathogenicity will have a child with Fragile X disorder, which has important counseling implications [16,17]. The importance of uninterrupted repeats, rather than repeat length *per se*, has been shown for several of the spinocerebellar ataxias, Huntington's Disease, myotonic dystrophy, and other repeat expansion disorders [18–24]. In these disorders, interruptions may modify the age of onset, the probability of transmission, and clinical expressivity. Collectively, these findings underscore the importance of developing accurate methods to determine not just allele lengths, but full repeat sequences with their motifs and interruptions.

There is an increasing number of long-read TR genotypers that attempt to capitalize on this rich data while taking different approaches to overcome its limitations. More than 20 long-read TR genotypers have been introduced since 2015 [25]. These tools have evolved alongside discoveries of new loci and mutation dynamics, extending beyond characterizing length to include sequence variation at the locus and methods to decompose sequences into component motifs. These tools have also improved their ability to phase variants and to model genotypes at loci with high somatic instability, while maintaining reasonable computational time and memory requirements. Despite this explosion in computational method development, little is known about the tools' sequence-level accuracy, particularly at longer alleles and VNTRs. Most benchmarking to date has been performed by the developers of specific tools, comparing their methods to previous ones, and has often focused on allele length rather than the full allele sequence, despite sequence composition being essential for interpreting motif structure, interruptions, and pathogenic associations. Thus, there is a major gap, especially as the field increasingly relies on LRS for clinical variant discovery, interpretation, and population-scale studies. As part of the 1000 Genomes Long-Read Sequencing Consortium (1KGP-LRSC) [26],

we systematically evaluated openly available TR genotyping methods using a diverse set of accuracy metrics, with the aim of informing tool selection and study design for future large-scale long-read TR analyses, including the development of public reference resources.

We conducted a comprehensive sequence-level benchmark of seven actively maintained long-read TR genotypers selected for their broad adoption: STRkit [27], LongTR [28], Straglr [29], STRdust [30], vamos [31], ATaRVa [32], and Medaka Tandem [33]. For the six tools that can report full allele sequences, we assessed both length and sequence accuracy. Straglr did not report sequences and was therefore assessed solely on length. For this benchmark, we used public Oxford Nanopore Technologies (ONT) whole-genome LRS data from more than 100 individuals generated by the Genome in a Bottle (GIAB) Consortium [34], the Human Pangenome Reference Consortium (HPRC) [35], and adaptive sampling data that targeted relevant genes of individuals with pathogenic expansions [36]. We genotyped 43,009 TR loci from the TReXplorer catalog [37] and STRchive, selected to span a wide range of motif lengths, repeat lengths, repeat complexity, and genomic contexts. While several benchmarking call sets have been developed [38–41], all methods to measure TR alleles have limitations, making a gold-standard “truth” for TR benchmarking challenging to define. Therefore, we benchmarked accuracy using four complementary strategies: concordance with haplotype-resolved, high-quality HPRC assemblies; Mendelian consistency in a GIAB trio; cross-tool agreement; and sensitivity to pathogenic alleles confirmed by repeat-primed PCR (RP-PCR) and Southern blot. We evaluated both allele length and full-sequence similarity using the Levenshtein distance and assessed installation, compute requirements, documentation completeness, and output formats to reflect real-world usability. We discuss their strengths not only in terms of accuracy and performance, but also in the information they provide for each locus to support downstream analysis, and critically, their usability, from installation to execution and output parsing. Finally, we discuss current limitations and challenges for users, highlighting areas for future development.

Results

Genotyper selection

We reviewed the literature and Long Read Tools resource [42,43] to identify 25 long-read TR genotyping algorithms (**Additional file 1: Table S1**, [27–33,44–60]). From these, we selected open-source tools that had at least one update since 11 November 2025 that accepted ONT sequencing reads and genotyped both STRs and VNTRs. We further narrowed our selection to genotypers that can output full allele sequences in Variant Call Format (VCF), allowing users to identify motif sequences, interruptions, and compound allele structures. For comparison, we included Straglr, the most popular length-only genotyper. The final selection included eight tools (**Additional file 1: Table S2**), and of these, we were able to install and run seven (**Table 1**, and see Usability): STRkit [27], LongTR [28], Straglr [29], STRdust [30], vamos [31], ATaRVa [32], and Medaka Tandem [33]. For some analyses, we ran two versions of vamos, v2.1.7 and v3.0.5, because a new version was released during the analysis phase of this study. Vamos v3.0.5 is used unless otherwise specified. We were unable to install TRsv [61] without

substantial code modification, so we removed it from the study. Note that TRGT [45] was excluded as its license prohibits use on non-PacBio data.

Table 1. Tools evaluated in this study with key distinguishing features.

Genotyper (version)	Full allele sequence in VCF	Distinguishing features	Language	Source code
STRkit v0.24.2 [27]	Yes	Visualization, Mendelian/ <i>de novo</i> mode	Python	https://github.com/davidlougheed/STRkit
LongTR v1.2 [28]	Yes	Long-read implementation of HipSTR	C++	https://github.com/gymrek-lab/LongTR
Medaka Tandem v2.1.1 [33]	Yes	Estimates error model from reads, polished consensus, not published	Python	https://github.com/nanoporetech/medaka/tree/master/medaka/tandem
STRdust v0.16.0 [30]	Yes	Visualization (pathSTR)	Rust	https://github.com/wdecoester/STRdust
ATaRVa v0.5.0 [32]	Yes	Motif decomposition	Python	https://github.com/SowpatiLab/ATaRVa
vamos v2.1.7/v3.0.5 [31]	Optional read consensus (-S)	Motif annotation in assemblies and reads	C	https://github.com/ChaissonLab/vamos
Straglr v1.5.5 [29]	No	<i>De novo</i> mode, featured in the EPI2ME wf-human-variation pipeline [62]	Python	https://github.com/bcgsc/straglr

Usability

Catalog generation

Unlike most modern SNV, short indel, and SV genotyping methods, TR genotyping is typically done at a pre-defined set of loci based on an annotation of the reference genome. The genotypers we assessed all require such a catalog, often in Browser Extensible Data (BED) format and with slightly varying formats (see **Methods**). Most use a 0-based standard first three BED columns: chromosome, start position, end position, followed by a fourth motif sequence column. LongTR uses 1-based start coordinates in the catalog, while vamos uses a 1-based coordinate system by default, but can accept 0-based with the -Z flag. Tools typically provide their own appropriately formatted catalog for GRCh38, often including known pathogenic loci

and/or broader genomic catalogs based on TRF annotations. To assess tools using a common catalog, we developed a benchmark catalog of 43,009 loci by sampling isolated loci from the TRExplorer catalog [37] (see **Methods**). As the human genome is disproportionately rich in smaller motifs, and to fully explore this metric, we designed the catalog to sample up to 1000 loci for each motif size, then supplemented it with the 75 disease-associated loci in STRchive [5].

Genotyper installation and implementation experience

All benchmarking workflows were implemented using the Nextflow workflow management system (DSL2), enabling scalable and reproducible execution across high-performance computing environments. Each Nextflow process was defined within a fully self-contained environment, avoiding reliance on inherited shell modules and preventing clashes between dependency versions, ensuring consistent behavior across compute nodes. Tool dependencies were managed using either Singularity container images or Conda environments, depending on software compatibility: LongTR, Straglr, and STRkit were deployed via containers (Wave/Sequera community images or GitHub Container Registry), while ATaRVa, Medaka, and vamos required Conda environments; STRdust was run as a statically pre-compiled binary.

During implementation, we encountered several practical software and compatibility issues that affected successful execution or required additional debugging before tools could be incorporated into the benchmarking workflow. For Medaka, we encountered two issues that we reported to the developers. First, when input CRAM files contained a UR tag in the header pointing to a reference path that was not present on the local filesystem, Medaka did not use the user-supplied reference and instead attempted to use the header-embedded path, causing the run to fail (issue #574); note that limited CRAM support is a known limitation class across genomics tools and is not specific to Medaka. Second, when Medaka tandem was run on certain GIAB samples (HG002 and HG006), it processed approximately 98% of loci and then failed to terminate cleanly, exhausting wall-time limits and leaving large numbers of temporary files behind (issue #583); this issue was addressed by the developers shortly after reporting. For vamos, the TRExplorer catalog could not be used directly because vamos does not support overlapping loci and raises a hard error when the start coordinate of one locus precedes the end coordinate of the previous locus. Notably, even the vamos-specific catalog exported from TRExplorer initially contained overlapping entries, requiring a corrected catalog from the TRExplorer team before benchmarking could proceed (issue #10).

VCF output formats

The usability of a tool depends largely on ease of interpretation and integration with downstream analyses. The VCF format, originally developed for reporting simple genomic variants, was extended in version 4.4 to include TR specifications [63]. However, this extension has not been widely adopted. Before this update, TR variant callers for SRS platforms relied on flexible adaptation of the standard VCF format, and these conventions were largely carried forward by long-read genotypers. As a result, output formats vary across tools, sometimes dramatically.

Most tools report VCF results, although some also provide auxiliary output files. Straglr generates a tab-separated values (TSV) file containing read-level information for each locus and a summary variants file in BED format. STRkit outputs an overview in TSV format, along with a more extensive JavaScript Object Notation (JSON) report. Medaka Tandem produces multiple supplementary files, including reports of missing regions, consensus allele sequences, and binary alignment map (BAM) files of reads mapped to consensus alleles.

LongTR, ATaRVa, STRkit, Medaka Tandem, and STRdust all produce VCF output with complete allele sequences in REF and ALT fields, while using distinct keys in FORMAT and INFO fields. Straglr instead reports allele lengths in the INFO field. vamos includes most relevant information within the VCF with the addition of the -S parameter, but its formatting differs from that of other tools and generally requires additional reformatting for compatibility (see **Methods**).

Computational Resources

Computational efficiency is becoming increasingly critical as analyses move to population scales in cloud compute environments, where even modest differences in resource requirements can be costly. We developed a Nextflow [64] pipeline to run all genotypers and record computational resource usage (<https://github.com/dashnowlab/TR-Benchmarking>) [65,66]. Tools were run on one of the University of Colorado Boulder Alpine's [67] x86_64 AMD Milan CPU in single-thread mode with 32GB of physical memory, which was increased to 64GB for failed runs (Straglr only) (**Additional file 1: Table S3**). We attempted to run all genotypers on the ~5 million loci in the full TRExplorer Catalog; however, Alpine limits jobs to one week, and some tools were still running after seven days. We therefore focused our subsequent benchmarking efforts on the described curated ~43k TR catalog. Physical memory usage was modest for most tools (**Figure 1A**): 1-1.5 GB per sample for vamos, ATaRVa, and STRdust. STRkit, LongTR, and Medaka Tandem typically required 2.5-3.5 GB, with LongTR using up to 6 GB for some samples. Straglr had a much higher mean memory usage of 37.3 GB, with up to 70 GB for one sample. Vamos was the fastest (mean 73 mins, **Figure 1B**), closely followed by STRkit (90 mins) and LongTR (98 mins). Runtimes for ATaRVa, Medaka Tandem, and STRdust were typically around 200 minutes. Straglr's mean runtime was 13.5 hours per sample, with a maximum of almost 20 hours. All tools can be run with multi-threading to substantially reduce elapsed runtime, except for LongTR, which can be run by chromosome or other subsets, followed by merging, but this must be handled by the user.

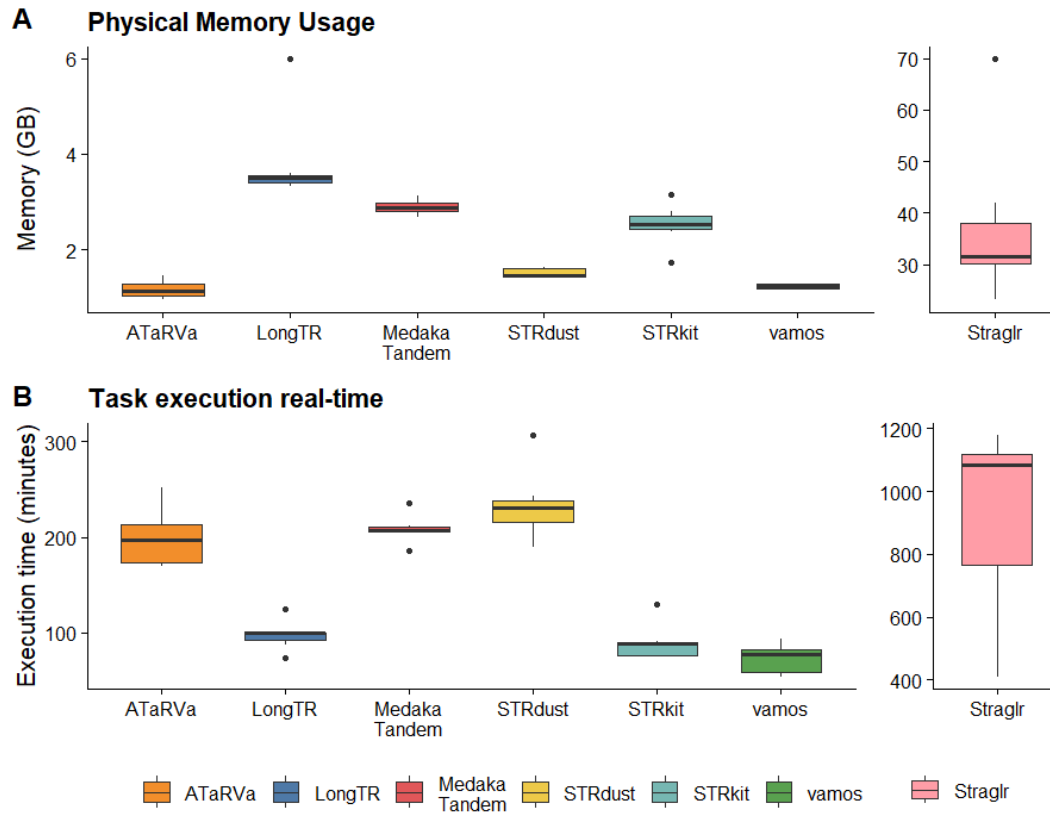


Figure 1. The seven GIAB samples were subsampled to ~30X and genotyped at 43,009 TR loci. Tools were run on one of CU Boulder Alpine's x86_64 AMD Milan CPU in single-thread mode with 32GB of physical memory. STRaglr failed at 32 GB of memory and re-run with 64GB of memory. **A.** Memory describes the peak physical RAM (peak resident set size) per sample, while **B.** Execution time is the elapsed run time for the genotyping task on a single thread.

Accuracy

We devised four strategies to assess accuracy: assembly concordance, Mendelian consistency, genotyper consistency, and sensitivity to detect pathogenic expansions confirmed by orthogonal molecular techniques (**Figure 2**). Two comparison metrics were used: 1) we compared total allele length in bp, and 2) assessed full sequence similarity using Levenshtein distance to calculate the number of edits between two alleles.

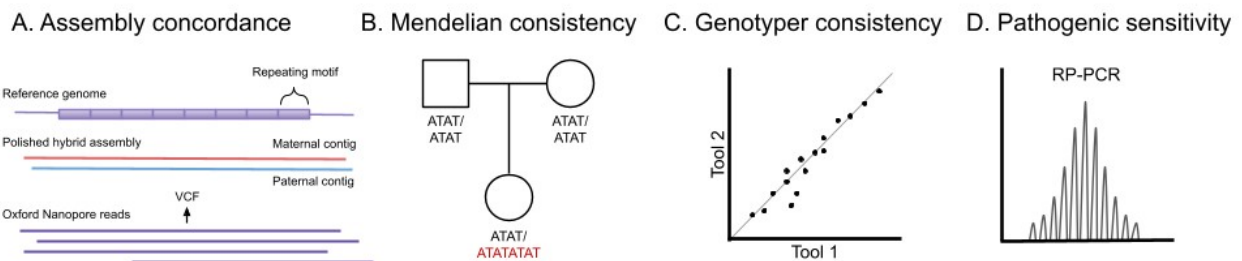


Figure 2. Genotyping tools were run on 30-35x ONT reads aligned to GRCh38 to assess assembly

concordance, Mendelian consistency, genotyper consistency, and sensitivity to pathogenic expansions. Two comparison metrics were used: 1) length, and 2) full sequence using Levenshtein distance. **A.** Polished hybrid ONT/HiFi assemblies from HPRC were aligned to **GRCh38** to assess assembly concordance. For each tool's VCF record, we extracted the corresponding alleles from the assembly. **B.** In the two GIAB trios, probands were compared to parents at each locus to assess Mendelian consistency. Any Mendelian inconsistency (i.e., apparent *de novo* mutation) was assumed to be an error. **C.** For each pair of tools, genotyper consistency was assessed using allele similarity at loci called by both tools. **D.** Tools were run on data collected using adaptive sampling on the ONT platform from samples with known pathogenic alleles confirmed by repeat-primed PCR (RP-PCR) and, in some cases, Southern blot. Results were then assessed for sensitivity based on their ability to identify an allele in the disease pathogenic range.

Assembly concordance

We benchmarked read-based genotyping against assemblies using data for 67 HPRC samples with equal mix sequenced using the R9.4.1 chemistry (n=33) and the R10.4.1 chemistry (n=34). We genotyped the catalog loci using all seven tools on GRCh38-aligned ONT read data and, for comparison, assembly genotypes are extracted using an in-house script on GRCh38-aligned assemblies (see Methods). Note that some tools (STRkit, LongTR) extend the coordinates from those in the catalog; therefore, we chose not to simply extract the catalog coordinate from the assembly and use a method such as Truvari [68].

For each genotype, alleles called by each tool were paired with assembly alleles such that the combined deviation or error was minimized. For each allele, the absolute deviation from the corresponding assembly allele length was calculated and categorized into four classes: *Match* (no error), *Off by 1bp* (error of exactly 1 bp), *Off by motif* (error less than or equal to the motif length of the locus), and *Mismatch* (error exceeded the motif length) (**Additional file 2: Fig. S1**). LongTR applies a default minimum mean base Phred score threshold of 30 per read and had no calls in samples sequenced with R9.4.1 chemistry, as the mean read base quality for these samples falls below this threshold. The concordance results are thus presented separately for the R10.4.1 and R9.4.1 samples.

While all tools were provided with the same ~43k TR catalog, most genotypers called a mean of ~42k loci per sample except Straglr (R10: 36,966 loci; R9: 38,019 loci) and STRkit (R10: 29,788 loci; R9: 30,011 loci) (**Additional file 1: Table S4**). The missing calls in STRkit could be attributed to a normalized map quality threshold of 0.9, which filters out many supporting reads.

We calculate concordance for a tool as the percentage of called loci that fall into the deviation classes Match and Off by 1 bp in a sample. The mean concordance of the tools, ranked from best to worst by R10 concordance was: LongTR (R10: 98.67%), Medaka Tandem (R10: 98.61%; R9: 97.77%), ATaRVa (R10: 98.48%; R9: 97.35%), vamos (R10: 98.09%; R9: 95.94%), and STRkit (R10: 97.72%; R9: 96.21%) (**Figure 3A**). Beyond these five tools, there is a notable drop in concordance with Straglr (R10: 90.09%; R9: 88.22%) and STRdust (R10: 72.56%; R9: 63.56%) (**Figure 3A**). However, when an off-by-motif deviation is permitted, the concordance of STRdust increases to 94.93% (**Additional file 1: Table S5**).

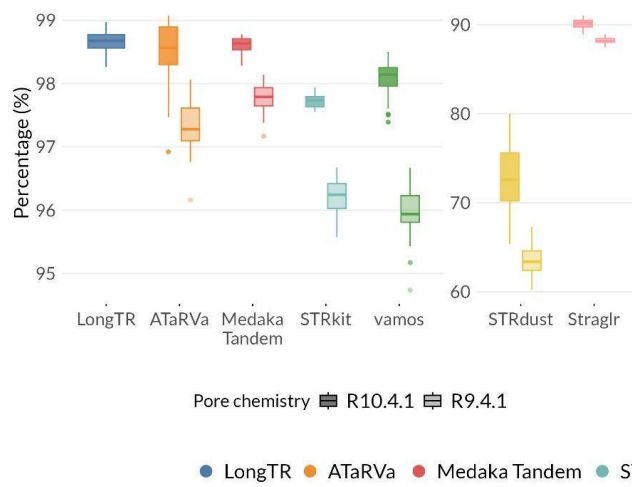
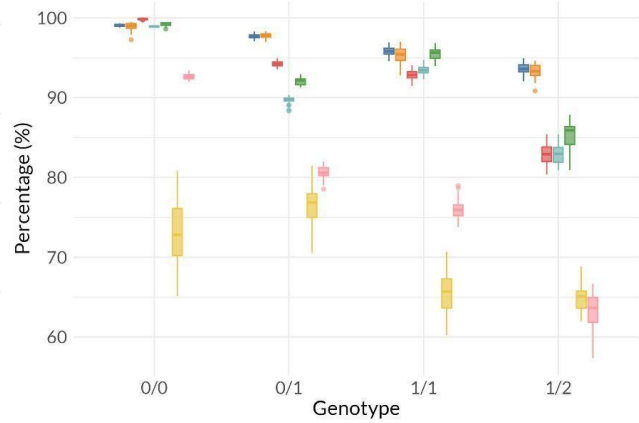
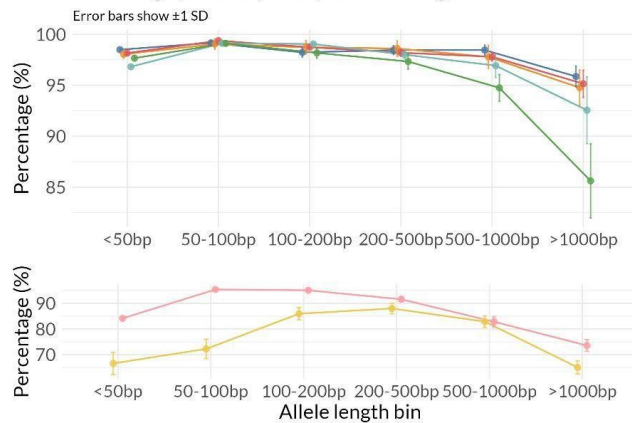
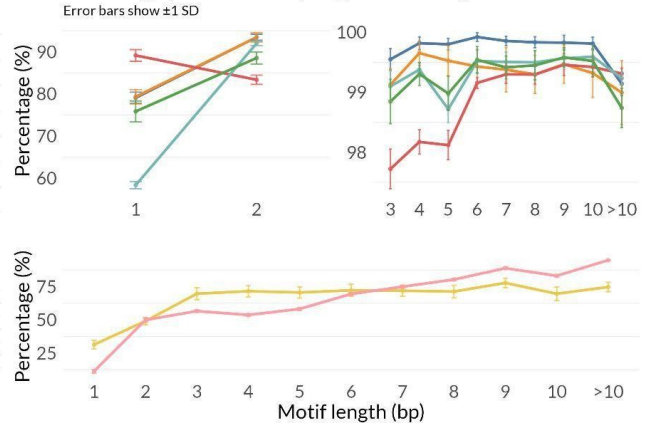
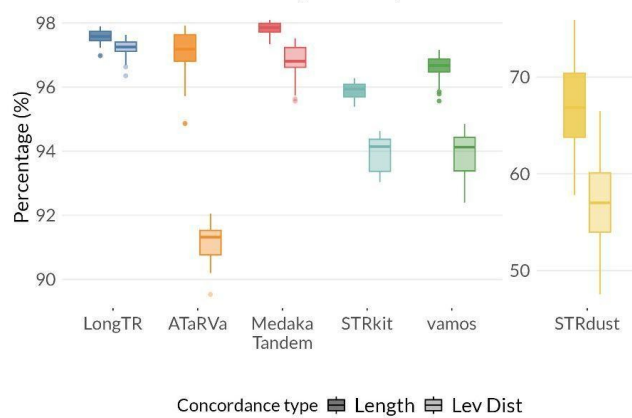
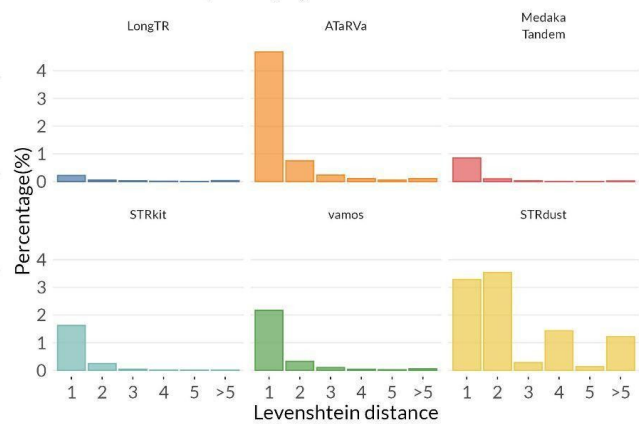
A Percentage Concordance (Match & Off by 1bp)**B** Percentage Concordance (Match & Off by 1bp) vs Genotype**C** Percentage (<= Off by motif) vs Allele Length Bin**D** Percentage (Match & Off by 1bp) vs Motif length**E** Concordance based on Length and Sequence**F** Concordance drop at varying levenshtein distances

Figure 3: Assembly concordance of genotypers across different stratifications for R10 samples. **A.** Box plots show the distribution of concordance for each tool, across all samples stratified by pore chemistry (R10.4.1, solid boxes; R9.4.1, hatched boxes). Tools are split across two y-axes for readability. **B.** Concordance of each tool stratified by genotype class: homozygous reference (0/0), heterozygous reference (0/1), homozygous alternate (1/1), and heterozygous alternate (1/2). Points represent mean concordance per genotype class across all samples. Points represent mean concordance, and error bars indicate ± 1 standard deviation across samples. **C.** Concordance of each tool across allele length bins (<50 bp, 50–100 bp, 100–200 bp, 200–500 bp, 500–1000 bp, and >1000 bp) in R10.4.1 samples. Points represent mean concordance, and error bars indicate ± 1 standard deviation across samples. Tools are displayed in two separate panels for readability. **D.** Concordance of each tool across motif length categories (1–10 bp and >10 bp) in R10.4.1 samples. Points represent mean concordance, and error bars indicate ± 1 standard deviation across samples. Tools are displayed across three panels for readability. The upper two panels show the five higher-performing tools (LongTR, ATaRVa, Medaka Tandem, STRkit, and vamos), with mononucleotide and dinucleotide repeats shown separately on a y-axis range of 50–95% and motif lengths ≥ 3 bp displayed on an expanded y-axis of 98–100%. The lower panel shows STRdust and Straglr on a broader scale to capture their wider concordance range across all motif lengths. **E.** Concordance calculated for each tool on two levels: length-based concordance, considering all calls with a matching allele length, and sequence-based concordance, calculated using Levenshtein distance for the subset of calls with a matching allele length. The difference between the two values reflects the proportion of length-concordant calls that differ at the sequence level from the assembly allele. **F.** Contribution of Levenshtein distance categories to percentage-point sequence-level concordance drop. Bar plots showing the percentage of total calls at each Levenshtein distance value for each tool, restricted to calls with matching allele lengths. Each bar represents the proportion of calls at a given Levenshtein distance, illustrating the relative contribution of different magnitudes of sequence deviation to the overall sequence-level concordance drop observed for each tool.

On average, 83.6% of loci in the R10 HPRC assemblies were invariant (homozygous for the reference allele). To assess tool performance at variable loci, we excluded loci with homozygous reference calls in assemblies. STRdust exhibited the smallest percentage-point drop in concordance between all loci and non-reference loci at just 1.42%; however, this should be interpreted in the context of its lower overall concordance relative to other tools. LongTR and ATaRVa showed near-identical drops of 2.15% and 2.17%, respectively, followed by progressively larger drops in vamos (5.91%), Medaka Tandem (6.43%), STRkit (7.69%), and Straglr (13.47%) (**Additional file 2: Fig. S2**).

To further characterize tool performance, we categorized variant loci by genotype: heterozygous reference (0/1), homozygous alternate (1/1), and heterozygous alternate (1/2), and assessed each tool's performance within these groups. In the assemblies, the average proportions of genotype calls are 83.70% (0/0), 8.71% (0/1), 5.21% (1/1), and 2.36% (1/2). Most tools show broadly similar genotype proportions to the assemblies, with a few exceptions (**Additional file 2: Fig. S3**). STRdust tends to call more genotypes with an alternate allele, while ATaRVa and Straglr show slightly elevated 0/0 call rates (88.81% and 88.21%, respectively). STRkit, Medaka Tandem, and Straglr have fewer heterozygous (0/1 and 1/2) calls than assembly, with correspondingly higher 1/1 calls, suggesting a slight bias toward homozygous calls.

Across all tools, concordance consistently decreases with increasing numbers of non-reference alleles (**Figure 3B**). STRkit and vamos are exceptions, showing higher concordance for 1/1 than 0/1 genotypes, potentially reflecting a homozygous calling bias. LongTR and ATaRVa are the most robust, maintaining concordance above 93% even for 1/2 calls. For 0/0 genotypes, all tools except STRdust achieved >90% concordance, with five tools approaching 98%. The per-

genotype ranking broadly mirrors the overall concordance order of LongTR, ATaRVa, Medaka Tandem, vamos, STRkit, Straglr, and STRdust, with two notable exceptions: Medaka Tandem was the top performer for 0/0 genotypes (99.83%), and ATaRVa for 0/1 genotypes (97.76%).

We next assessed tool concordance as a function of total allele length, binning alleles into six categories: <50 bp, 50–100 bp, 100–200 bp, 200–500 bp, 500–1000 bp, and >1000 bp. Concordance generally decreased with increasing allele length for loci above 100 bp, consistent with the expectation that longer alleles are likely to accumulate more sequencing errors (**Figure 3C**). All tools also showed poorer concordance for loci below 50 bp compared to the 50-100bp bin, likely because this bin contains a disproportionate number of homopolymers. Examining the number of calls per category (**Additional file 2: Fig. S4**), STRkit had the fewest calls across all length categories, with a proportionally increasing rate of missing calls at greater lengths. In contrast, despite having the second-highest overall rate of missing calls, Straglr concentrated most of its missing calls in the shortest length category and had the highest number of calls for alleles >1000 bp. All tools produced call counts comparable to the assemblies across most length categories, with the notable exception of the >1000 bp category, which tools tended to under-call. Among tools, Straglr, STRdust, and vamos had the highest number of calls in this longest category. When considering the absolute count with accurate calls across all tools and length categories, performance was largely comparable, with Straglr showing a slight advantage in the >1000 bp category (**Additional file 2: Fig. S5**).

While above we considered total allele size, we also evaluated each tool's ability to detect expansions relative to the reference across a range of size categories: <50 bp, 50–100 bp, 100–200 bp, 200–500 bp, and >500 bp (**Additional file 2: Fig. S6**). All tools showed a gradual decline in concordance with increasing expansion length, with a more pronounced drop for expansions >1000 bp. A consistent dip in concordance was also observed across all tools in the 100–200 bp bin. ATaRVa and LongTR were the top two tools overall, with ATaRVa performing the best up to 500 bp expansions and LongTR performing better beyond that threshold. For the largest expansion category (>1000 bp), LongTR achieved the highest concordance at 89.66% while STRkit had the lowest at 24.37%.

We next evaluated tool performance across motif length categories. Recall that the catalog was constructed by sampling 1,000 loci per motif length. The total number of calls per category was broadly consistent across tools, except for Straglr and STRkit. Straglr showed a proportionally uniform distribution of missing calls across motif length categories, whereas STRkit showed a progressively increasing rate of missing calls with longer motifs (**Additional file 2: Fig. S7**).

When comparing concordance across tools, a clear split emerges: STRdust and Straglr perform markedly worse than the remaining five tools across motif lengths (**Figure 3D**). Mononucleotide (homopolymer) and dinucleotide repeats posed the greatest challenge across all tools, likely reflecting high read-level error or possibly mosaicism at these repeat types. Notably, Medaka Tandem achieved the highest concordance for homopolymers at 89.13%, nearly 10% better than the next best tools, ATaRVa (79.31%) and LongTR (79.07%), and Straglr showed the poorest concordance at 11.34%. (**Figure 3D**). However, Medaka Tandem dropped to the lowest concordance among the top five tools for dinucleotide repeats (83.25%), while the other four

tools performed better (88–93.5%), with LongTR and ATaRVa performing the best. From motif length 3 bp onward, all five tools converge to accuracies of 98% or above, remaining largely stable through longer motif lengths before a slight decline in the >10 bp category. Both STRdust and Straglr improve with increasing motif length, though their trajectories differ (**Figure 3D**).

Beyond length-based concordance, we assessed each tool's sequence-level concordance by comparing tool allele sequences against the assembly using Levenshtein distance. Straglr was excluded from this analysis as it does not report allele sequences. To isolate sequence-level deviations, we restricted the analysis to loci where the tool length and assembly length matched, then compared the proportion of calls with matching sequences (Levenshtein distance of 0, **Figure 3E**). LongTR demonstrated the greatest sequence-level robustness, achieving 97.20% sequence concordance with a drop of only 0.36% from its length-based concordance. Medaka Tandem was the next-best performer, with a drop of 1.01%, followed closely by STRkit (1.95%) and vamos (2.72%). ATaRVa and STRdust showed notably larger deviations, with drops of 5.92% and 9.89%, respectively. Examining the distribution of Levenshtein distances at length-concordant loci, the concordance drop for most tools is largely explained by calls with a Levenshtein distance from the assembly of 1 (i.e., a single bp SNV, **Figure 3F**). STRdust is a notable exception as it generated calls with larger Levenshtein distances, suggesting a higher prevalence of more substantial sequence-level errors.

When analyses were repeated using R9.4.1 data (except for LongTR), all tools experienced a drop in performance, though the overall ranking trends were largely preserved across metrics. ATaRVa and Medaka Tandem competed closely for the top positions, with each taking the lead in different analyses, while vamos and STRkit similarly traded places in different analyses, followed by Straglr and STRdust (**Additional file 2: Fig. S8**). One notable exception was STRkit, which outperformed all other tools for alleles >1000 bp in R9 data (81.83%; **Additional file 2: Fig. S7B**), in contrast to its fourth-place ranking in the R10 analysis.

Mendelian consistency

To assess the accuracy of TR genotypes produced by each tool, we evaluated Mendelian consistency in a parent-offspring trio, reasoning that concordance with expected inheritance patterns provides an orthogonal measure of genotyping fidelity independent of the reference genome. In the HG002 trio, the proband was compared to parents at each genotyped locus, and any Mendelian inconsistency (i.e., apparent *de novo* mutation) was assumed to be an error. Because the *de novo* mutation (DNM) rate for TRs is estimated to be on the order of 10^{-5} per locus per generation [1–4], we would expect fewer than 1 DNM per generation from our catalog of 43,009 loci, meaning that the vast majority of apparent DNMs are likely to be errors.

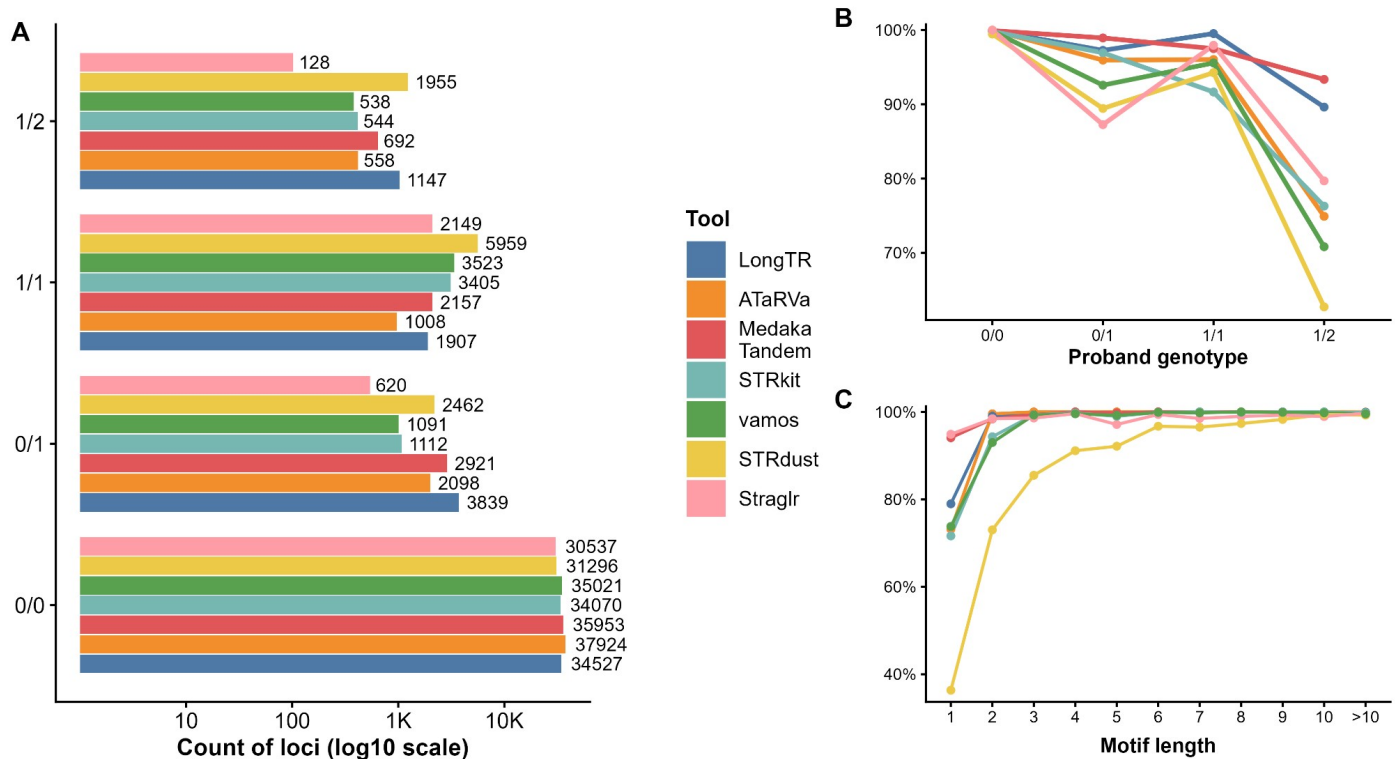


Figure 4. Mendelian consistency and motif-length performance across STR genotyping tools in the HG002 trio. **A)** Number of genotyped loci across four child genotype classes (0/0, 0/1, 1/1, and 1/2) for each tool. Counts are shown on a log10 scale. Values next to bars indicate the total number of successfully genotyped loci per genotype and tool. **B)** Mendelian-consistency rate (percentage of loci where the proband alleles are within one motif length of the parents) stratified by child genotype. **C)** Mendelian-consistent rate as a function of STR motif length. Motif lengths greater than 10 bp are collapsed into a single bin.

We evaluated Mendelian consistency in the GIAB HG002 trio by assessing whether the child's genotype could be explained by parental alleles, allowing for exact allele matches, calls differing by one base pair, or calls differing by up to one full repeat motif unit; a composite metric that accounts for the rounding and boundary ambiguities inherent to TR genotyping. Loci were stratified by the child's genotype class (0/0, 0/1, 1/1, and 1/2) and by repeat motif length.

The total number of genotyped loci varied substantially across tools, which is a critical consideration when interpreting consistency metrics. Several tools genotyped similar numbers of loci (~41.5k; Medaka Tandem, ATaRVa, LongTR, and STRdust), with *vamos* (n=40,173) and STRkit (n=39,131) calling slightly fewer, and Straglr (n=33,434) notably fewer. The difference is particularly pronounced at compound heterozygous loci (1/2), where STRdust called the most (n=1,955), followed by LongTR (n=1,147), Medaka Tandem (n=692), ATaRVa (n=558), STRkit (n=544), and *vamos* (n=538), while Straglr called only 128. This means Straglr's consistency estimates at 1/2 loci are based on a much smaller and potentially less representative set, and its apparent accuracy may partly reflect more selective genotyping (Figure 4A).

At homozygous reference loci (0/0), nearly all tools achieved >99.8% consistency (ATaRVa, Straglr: 99.99%; LongTR: 99.95%; Medaka Tandem: 99.93%; STRkit: 99.81%; vamos: 99.86%), indicating that false positive heterozygous calls are rare across methods. STRdust was the notable exception at 99.5%, driven by a higher rate of motif-length-off calls at this class (**Additional file 1: Table S6, Additional file 2: Fig. S9**). Consistency declined as genotype complexity increased, with the sharpest divergence among tools observed at compound heterozygous loci (1/2), which represent the most challenging class. Here, Medaka Tandem remained the top performer at 93.4%, followed by LongTR (89.6%), Straglr (79.7%), STRkit (76.3%), ATRaVa (74.9%), and vamos (70.8%), while STRdust showed the lowest consistency at 62.7% (**Figure 4B**). At heterozygous loci (0/1), Medaka Tandem again led (98.9%), followed by LongTR (97.3%), STRkit (96.9%), ATRaVa (96.0%), vamos (92.6%), STRdust (89.4%), and Straglr (87.3%). At homozygous alternate loci (1/1), LongTR achieved the highest consistency (99.5%), followed by Straglr (97.9%), Medaka Tandem (97.5%), ATRaVa (96.0%), vamos (95.6%), STRdust (94.3%), and STRkit (91.7%).

Mendelian consistency also varied by repeat motif length (**Figure 4C, Additional file 1: Table S7**). Mononucleotide repeats were the most difficult to accurately call, with STRdust below 40% consistency, and ATRaVa, STRkit, and vamos ranging from 71% to 74%, reflecting the well-known higher homopolymer error in LRS. Medaka Tandem (94.1%) and Straglr (94.9%) performed notably better at mononucleotide loci. Performance improved sharply for di- and trinucleotide repeats and largely plateaued at ≥4 bp motifs, where most tools converged on 95–100% consistency. STRdust showed the steepest improvement trajectory across motif lengths, rising from below 40% at mononucleotides to above 99% at longer motifs, yet its overall consistency remains constrained by its weaker performance on shorter motifs and heterozygous genotypes.

Overall, Medaka Tandem demonstrated the most robust Mendelian consistency across all genotype classes and motif lengths in the HG002 trio, with LongTR as a strong second. Straglr performed competitively, particularly at 1/2 and 1/1 loci, though its smaller genotyped set limits direct comparison. ATRaVa, STRkit, and vamos formed a competitive intermediate tier at most genotype classes, while STRdust showed the most tool-specific weaknesses, particularly at heterozygous and compound heterozygous loci and short motifs despite genotyping the broadest set of loci. To summarize performance across genotype classes without allowing the numerically dominant 0/0 class to dominate, we computed a simple unweighted average of total consistency across all four genotype classes (**Additional file 1: Table S6**). This confirmed the class-by-class interpretation: Medaka Tandem ranked first (97.4%), followed by LongTR (96.6%), with ATRaVa (91.7%), Straglr (91.2%), and STRkit (91.2%) forming a close intermediate group, and vamos (89.7%) and STRdust (86.5%) trailing.

Genotyper consistency

Consistency across different genotyping methods does not necessarily reflect accuracy; however, such comparisons are useful to the community for selecting a genotyping method and for assessing the usefulness of a control cohort genotyped with a different method. Consistency across callers can also point to specific loci that are potentially problematic, owing to high levels of disagreement, and may help diagnose the causes of error. Additionally, when running

multiple genotypers to improve sensitivity, consistency analysis may inform the selection of complementary tools.

For each pair of tools, we assessed the similarity of allele sequences and lengths at the set of loci called by both tools. For comparisons with STRaglr, only length was considered. Genome assemblies are treated as an additional genotyping method, with genotype calls derived from the assembly based on the original catalog coordinates. For each pair of tools, we calculated the Levenshtein distance and the absolute length difference between alleles across all commonly called loci within a sample, and averaged these values across all loci and samples (**Additional file 2: Fig. S10**). The pairwise average values, including comparisons with assembly calls, are visualized as a heatmap (**Figure 5**), where the upper triangle shows the average Levenshtein distance, and the lower triangle represents average absolute length difference. Examining the average length difference first, LongTR, ATaRVa, Medaka Tandem, and vamos show high similarity with average length differences below 1. The increased average length difference observed between STRkit and the assembly, despite STRkit's high overall assembly concordance, is attributable to STRkit adjusting locus coordinates, while the comparisons here are done based on original catalog coordinates. Consistent with the overall assembly concordance results, Straglr shows the second-lowest deviation from the assembly, while STRdust shows the highest. When the assembly is included as a comparator, the average distance of each tool from the assembly is broadly comparable to, and in most cases slightly lower than, pairwise tool-to-tool distances. Notably, ATaRVa shows the smallest average length difference from the assembly, despite ranking third in overall assembly concordance. The average Levenshtein distance shows trends largely similar to those of the average length difference.

We next examined loci where the majority of tools disagreed with the assembly calls. Using one representative sample (HG00290), we selected all loci where at least five tools disagreed with the assembly (discordant loci: 402) and visualized pairwise tool comparisons restricted to these loci as a heatmap (**Additional file 2: Fig. S11**). At these discordant loci, both sequence and length differences between tool pairs are substantially smaller than the differences observed between individual tools and the assembly. This suggests that either all tools are systematically missing the assembly-based genotype, or that the read data presented to the tools differs from the sequencing data underlying the assembly, leading all tools to converge on the majority genotype supported by the available reads.

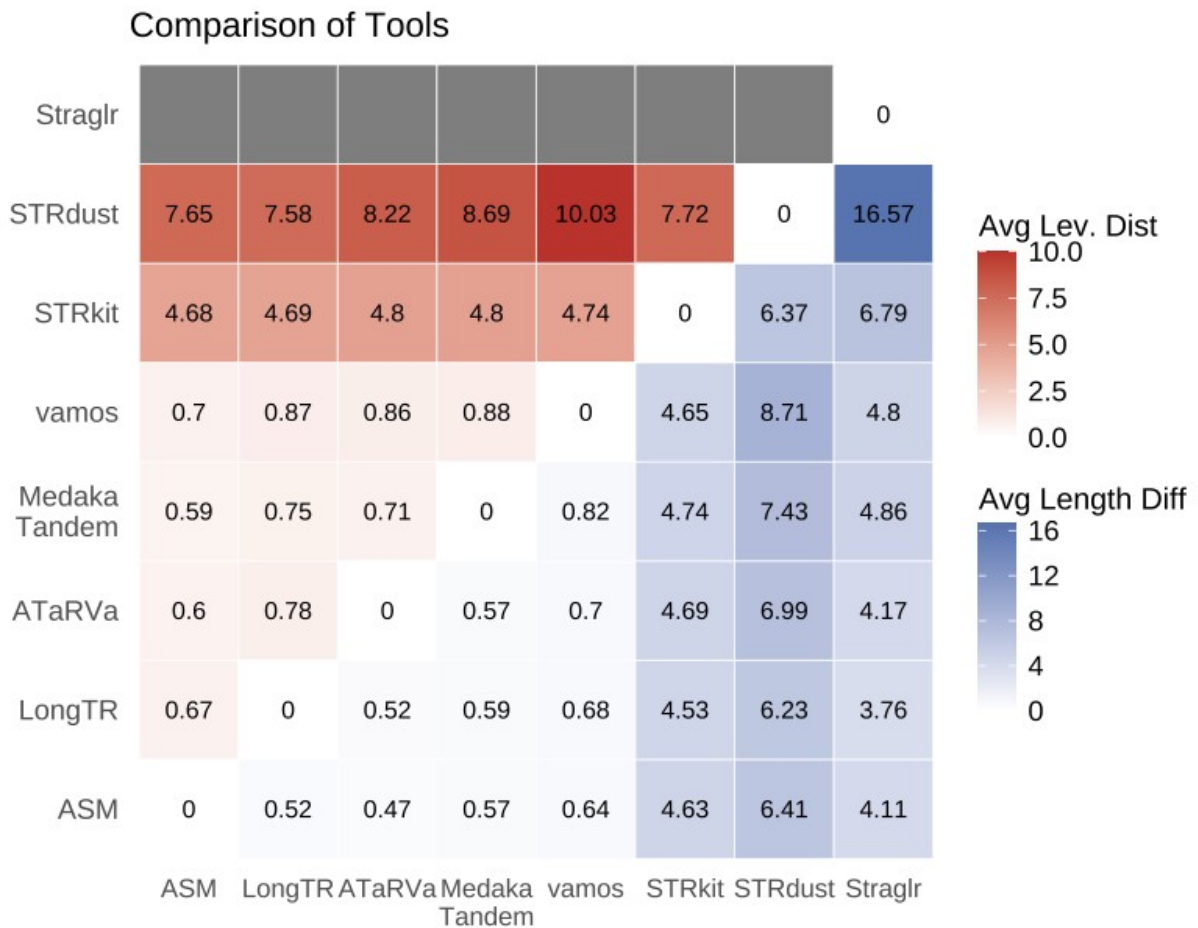


Figure 5. Genotyper consistency: pairwise comparison of genotyping tools based on allele sequence and length differences. The heatmap displays pairwise comparisons between all tools, including genome assemblies as a reference genotyping method. The upper triangle represents the mean Levenshtein distance between allele sequences, and the lower triangle represents the mean absolute length difference between alleles, averaged across samples at only loci genotyped by all genotypers. Straglr is excluded from sequence-level comparisons as it does not report allele sequences.

Sensitivity to Pathogenic Expansions

To assess sensitivity to pathogenic expansions, we ran all tools on blood-derived sequencing of individuals with a known TR disorder and/or an orthogonal molecular test. This previously published study [36] generated targeted ONT reads using adaptive sampling of 37 target genes with known TR expansions. Of those, nine genes had pathogenic expansions in this cohort: *ATXN1*, *C9orf72*, *DMPK*, *FMR1*, *FXN*, *HTT*, *NOTCH2NLC*, *PABPN1*, and *RFC1*. We analyzed 27 individuals who had known TR disorder and/or allele-size estimates from molecular testing

using RP-PCR, Southern blot, or both (**Additional file 1: Table S8**). The cohort included 19 individuals with a molecularly confirmed pathogenic TR expansion, 2 individuals with premutations, 2 carriers, and 3 unaffected family members without expansions. An additional sample was listed as *RFC1* positive and clinically affected, but no molecular testing results were available.

Reads from these individuals were aligned to GRCh38 and genotyped with all 7 tools using the same pipeline as above. We then subset the results to only the relevant disease locus for each sample. LongTR did not genotype any known TR disease loci in any of the samples because it requires at least 10 spanning reads, each with a mean base quality of 30. As this is R9 ONT data published in 2022, the mean base qualities per sample were 19.5 (range 18.5-20.8), and no disease loci passed these filters. Other tools genotyped most loci; however, because tools differed in their default read-depth requirements, to maximize sensitivity, we reran all tools and, where possible, adjusted parameters to call any locus with at least one read. The difference was most notable for LongTR (0 to 87%, **Figure 6A**), but most tools increased in sensitivity with these setting adjustments. The exception was Straglr, which detected an additional pathogenic expansion in default settings than in “min reads” settings. The initial analysis was performed with *vamos* v2.1.7; however, v3.0.5 was released during the analysis phase of this study. We noted a dramatic drop in sensitivity between versions from 61% to 30%, with the missed loci not included in the VCF. v2.1.7 was used for the remainder of the pathogenic loci analysis, and the potential bug has been reported. *Vamos* changed the read consensus algorithm between versions from abPOA to sPOA. *Vamos* v3.0.5 results are retained here for transparency and consistency with the assembly and Mendelian analyses, where it performed well.

None of the tools achieved 100% sensitivity (**Figure 6B**), and all missed the *C9orf72* expansion. There was read evidence of the expansion, but the aligner softclipped these reads, so none fully spanned the expansion (**Additional file 2: Fig. S12**). STRdust demonstrated consistently superior sensitivity across settings (91% in the default setting and 96% in the “min reads” setting). While the default settings of LongTR meant it did not call variants at any of these loci due to the lower quality of R9 data, it performed almost as well as STRdust after parameter adjustment (87%).

For long pathogenic alleles such as *FMR1* and *RFC1*, RP-PCR is known to dramatically underestimate allele sizes. We observed much longer sequence-based size estimates for these loci (**Figure 6C**), demonstrating the value of sequencing for sizing large expansions. The genotypers also identified the expected primary motif for all pathogenic loci, although the allele sequences contained numerous interruptions, likely due to high per-base error rates in the sequencing data.



Figure 6: Sensitivity to and sizing of pathogenic expansions. Samples were sequenced with adaptive sampling [36]. Where possible, tools were run with the most sensitive settings to include lower-quality reads, except for Straglr, which was run in default settings. vamos v2.1.7 was used. **A:** Most tools were more sensitive to pathogenic expansions when settings were adjusted to include lower-quality reads. Relevant settings were identified in all tools other than vamos, specifically: ATaRva (--min-reads 1), LongTR (--min-reads 1 --min-mean-qual 1), Straglr (--min_support 1 --min_cluster_size 1), STRkit (--min-reads 1 --min-read-align-score 0.1), STRdust (--support 1), Medaka Tandem (--min_depth 1). LongTR made no calls in these samples when run with default settings, as no reads passed filters. **B.** Sensitivity to detect pathogenic or premutation alleles based on length. The “pathogenic min” column indicates the pathogenic allele threshold in motifs for each disease from STRchive. For sample CD00014 the FMR1 premutation size is given. The “molec” column indicates size estimates from RP-PCR or Southern blot. All other columns are the allele sizes in motifs estimated by each computational method. When multiple alleles are called they are separated by “/”, while cells with a single value indicate only one allele was reported. A cell is colored green if the estimated allele size exceeds the pathogenic min and the genotype is consistent with the mode of inheritance for that disease. Where possible, tool filters were adjusted to consider reads of all qualities and alignment scores. Unaffected individuals were excluded. Total sensitivity is the % of affected samples in which the tool correctly identified a pathogenic genotype. **C.** Length concordance between RP-PCR and Southern blot molecular assays and genotypes for samples with known pathogenic TR expansions and their unaffected family members.

Relative performance across key metrics

We present the consolidated relative performance of all tools on computational efficiency and accuracy, choosing a representative metric for each (**Figure 7**) to help users select a tool suited to their project goals.

Computational performance evaluated based on runtime and memory requirements shows all tools, except Straglr, ran within configurations typical of personal computers. Vamos stood out with the most favorable resource usage across both metrics.

Tool accuracy was evaluated across three measures: concordance with high-depth assemblies, Mendelian concordance, and sensitivity to pathogenic expansions (**Figure 7**). In assembly concordance ATaRVa, Medaka Tandem, and vamos exceeded 99% concordance. Homopolymer repeats were a consistent challenge across all tools and a major source of inaccuracy, accounting for nearly half of all incorrect calls in the top-performing tools. Medaka Tandem performed substantially better than other tools at homopolymer loci. Identifying large expansions relative to the reference was another challenge for many genotypers. For loci with expansions exceeding 500 bp, LongTR performed best, followed closely by ATaRVa and vamos. For Mendelian consistency, Straglr, Medaka Tandem, and ATaRVa all exceeded 99%. A Mendelian consistency of 99% would imply approximately 300-400 apparent *de novo* variants, likely reflecting genotyping error rather than true *de novo* variation. STRdust emerged as the most sensitive tool for pathogenic expansions, missing a only single expansion call, despite its comparatively lower accuracy in genome-wide benchmarking. LongTR stands to be the second-best tool in this category.



Figure 7: Relative performance heatmap across key metrics of TR genotyping tools. Cell values show the observed metric, and colors represent performance scaled within each metric (0–1), with darker green indicating

better relative performance and yellow indicating worse relative performance. Usability metrics were based on the ~43k TR catalog in GIAB samples, single-threaded (see Methods). Assembly accuracy metrics comprise ≤ 1 motif unit consistency, homopolymer consistency, and sensitivity for expansions > 500 bp. Mendelian consistency represents the proportion of loci consistent with parental genotypes (MATCH + ONE-OFF; ≤ 1 motif unit difference) across all child genotypes in the HG002 trio. Pathogenic sensitivity: percentage of pathogenic or premutation genotypes identified with the best settings or version. * 61% for vamos v2.1.7, 30% for vamos v3.0.5.

Discussion

In this study, we conducted the first comprehensive sequence-level benchmark of long-read TR genotypers in Oxford Nanopore data. We assess seven genotypers across 43,009 TR loci in more than 100 human genomes. While previous evaluations were typically performed in support of a specific new tool and focused primarily on allele length, we show that a sequence-level approach more accurately estimates accuracy and reveals sequence-level discrepancies, especially for long alleles, VNTRs, and non-reference alleles. By integrating four complementary accuracy metrics, assembly concordance, Mendelian consistency, cross-tool agreement, and sensitivity to confirmed pathogenic expansions, we provide the most complete assessment to date of the strengths and limitations of current long-read TR genotyping tools. We also provide feedback on the user experience, including tool installation, computational resource requirements, and documentation, giving potential users a more realistic idea of what they are getting themselves into when they select a genotyper. Our work closes a critical gap and provides guidance for selecting TR genotypers for population studies, mechanistic research, and clinical diagnostics.

Trade-offs Between Tools: Is There a “Best” Genotyper?

No single genotyper was universally optimal in our analysis reflecting the inherent challenges of TR genotyping, where error rates vary considerably by locus characteristics. We observed variability in tool performance across homopolymers, long alleles, heterozygous loci, and non-reference alleles. This reflects distinct algorithmic sensitivities and error modeling strategies (**Additional file 2: Supplementary Note**) of each tool. For example STRdust demonstrated the highest sensitivity at pathogenic expansions, but lower relative performance on X loci suggesting overfitting, consistent with its development focused on analyses of clinically relevant tandem repeat regions in ONT data [30]. Assembly concordance also differed by sequencing chemistry; while all algorithms performed better on higher-quality R10 data, the difference in performance between R9 and R10 was greater for some tools. Notably, Medaka Tandem likely performs better on R9 and at homopolymers because it adapts its error modeling on a per-sample basis, effectively handling high-error reads. LongTR strikes a good balance between accuracy on typical-sized alleles and sensitivity to pathogenic expansions, making it a promising general-purpose choice [62]. However, this result was achieved on R9 data only by reducing read quality and alignment thresholds. In all cases, our initial experimentation suggests that many tools could perform substantially better with further parameter tuning.

Computational performance varied widely, with dramatic differences in memory and run-time, but importantly, we did not observe a consistent trade-off between computational cost and accuracy. Many of the fastest tools were also among the most accurate. Consequently, the “best” tool is context-specific: clinical diagnostic prioritizes sensitivity to very large pathogenic

expansions, while population-scale studies require computationally efficient, scalable methods. While all studies are likely to benefit from complete and accurate allele sequences, studies focused on motif evolution will especially benefit from motif decomposition, which is supported by only a small number of tools (e.g., *vamos* and *ATaRVa*).

Strengths and Limitations of Each Benchmarking Approach

Each of our benchmarking strategies offers distinct and often complementary strengths and limitations, so interpreting tool performance requires balancing what each can reveal. Assembly concordance provides a high-quality, but not infallible, comparison point: we do not assume assemblies are the perfect truth, but rather treat them as deep-coverage, multi-technology, orthogonal gold standards that complement our ~30x read-based genotyping. Because most TR loci are invariant, tools biased toward calling the reference allele can still score high in assembly length concordance. Mendelian consistency has similar pitfalls: a genotyper that always outputs the reference allele at every locus would achieve perfect Mendelian consistency and still appear to perform relatively well against assemblies, while completely failing to detect pathogenic expansions. However, a tool with high Mendelian consistency may generate fewer false-positive *de novo* variants in a trio analysis. Cross-tool agreement likewise does not necessarily indicate correctness; however, patterns of agreement and disagreement can be informative. When multiple tools agree with each other but diverge from the assembly, this often indicates assembly errors or allelic dropout in the reads, whereas widespread disagreement typically highlights difficult or unstable loci. Finally, sensitivity to pathogenic alleles is the most clinically relevant measure, and our analyses show that tools may perform well under other benchmarking characteristics but still miss many pathogenic expansions, some of the most challenging alleles to genotype. Taken together, these complementary strategies expose different and often non-overlapping error modes, underscoring the importance of multidimensional evaluation rather than reliance on any single metric.

Practical implementation challenges may limit adoption

A major theme emerging from this work was the substantial effort required to install, configure, execute, and integrate several long-read TR genotypers into a reproducible benchmarking workflow. We did not perform a formal usability study using established software evaluation methods; rather, we documented implementation challenges encountered during this process. These included dependency and environment issues during installation, incomplete or outdated documentation, limited examples for workflow integration, cryptic error messages, tool-specific assumptions that were not clearly documented, unexpected transformations of genomic coordinates between input catalogs and output VCFs, and VCF outputs that required downstream correction or reconstruction.

These challenges were encountered by an experienced bioinformatics team working in a controlled benchmarking setting and, in several cases, required inspection of source code, repeated troubleshooting, or direct communication with tool developers before analysis could proceed. In some cases these issues directly affected reproducibility within our own workflow. We therefore expect that such implementation barriers may reduce accessibility and slow uptake for users in settings without substantial computational support. Nevertheless, our observations point to concrete areas where software engineering improvements could

substantially increase usability in practice, including more complete and up-to-date documentation, clearer error reporting, more robust handling of common input formats and edge cases, improved guidance on parameter selection, and greater standardization of output representations beyond the current VCF specification.

Parameter Optimization: Essential but Under-documented

Parameter optimization emerged as another critical but under-documented aspect of TR genotyping. For the majority of our benchmark analyses, we deliberately focused on default or recommended parameters so as to reflect real-world usage. However, many tools expose dozens of tunable parameters that can substantially affect accuracy, sensitivity, and runtime. In our preliminary explorations of pathogenic loci, we observed dramatic improvements in sensitivity for most tools when parameters were adjusted to include lower-quality and less well-mapped reads. This suggests that our benchmark likely underestimates the maximum achievable performance of several tools. Effective parameter tuning is likely essential for specific scenarios, including long (i.e., most pathogenic) alleles, loci with significant somatic mosaicism, and regions dominated by homopolymers. Such optimization is likely to be especially critical in clinical settings to ensure sensitivity. Yet, guidance on parameter selection is rarely provided in tool documentation, nor explanations of how different settings interact with sequencing chemistry, coverage, or other error profiles. Although comprehensive parameter optimization was beyond the scope of this benchmark, our findings suggest some starting points for pathogenic loci and a call for developers to prioritize more comprehensive guidance and best-practice recommendations.

Toward an Ideal TR Genotyper

Although several tools performed well across particular scenarios, no single genotyper emerged as universally optimal, underscoring the need for continued method development. An ideal TR genotyper would offer practical usability, easy installation, clear documentation, and a standardized, interpretable VCF format. This should be paired with robust sequence-level performance across STRs and VNTRs. It should accurately reconstruct full allele sequences, resolve complex multi-motif structures, and maintain high accuracy across long alleles, non-reference alleles, and homopolymer-rich regions. Computational efficiency, explicit modeling of sequencing error, and optional read-level information to assess somatic variation would further enhance utility. Features such as haplotype phasing, uncertainty estimates, and built-in motif decompositions would make such a tool broadly applicable to population and clinical studies. Our findings highlight the methodological and usability gaps that must be addressed to approach this ideal and provide clear priorities for our community to enhance TR genotypers.

Strengths and Limitations of This Study

This study has several strengths. It is the first large-scale, sequence-level benchmark of long-read TR genotypers, conducted independently of any single tool. It included multiple orthogonal accuracy metrics, pathogenic expansions, a catalog enriched for challenging loci, and an independent assessment of usability and computational performance supported by reproducible workflows. Nonetheless, several limitations should be noted. Several authors have contributed to tools benchmarked in this study, specifically ATaRVa, STRdust, and vamos. We attempted to mitigate some of this bias by contacting the authors of all the tools we benchmarked via their

respective GitHub repositories when we encountered issues. Another limitation of our approach is that we predominantly used default parameters, and our limited exploration of this space indicates that default settings may underestimate performance, especially for higher-error data and pathogenic alleles. We did not assess the impact of somatic mosaicism on tool performance, nor the ability of tools to detect it. We also did not benchmark special features specific to a small number of tools, such as motif decomposition. Finally, this benchmark represents a snapshot of a rapidly evolving field, and we hope many of the tool limitations we observed will be addressed in future versions.

Conclusions

Our results show that no single genotyper performs best across all loci, metrics, and conditions; however, several tools offer strong performance for specific applications, while others provide more consistent performance across metrics, providing researchers with a set of good options rather than a single perfect solution.

This benchmark advances the field in several important ways. It demonstrates that sequence-level accuracy allows for more differentiation between genotypers than length accuracy alone. It highlights conditions under which accuracy deteriorates across many tools: homopolymers, higher sequencing error rates, and large pathogenic expansions. The benchmark highlights key methodological and usability challenges and calls for further development of both the tools themselves and the documentation for optimizing them.

Beyond evaluating current methods, this work offers practical recommendations to guide tool selection based on study goals and provides fully reproducible workflows to support community reuse, transparency, and future method development. Together, these contributions lay a foundation for future large-scale TR analyses, including those undertaken by the 1000 Genomes Long-Read Sequencing Consortium (1KGP-LRSC), by providing empirical guidance on method performance rather than prescribing a single default method. By identifying strengths and limitations across tools and study contexts, this work supports informed genotyping across studies of TR mutation dynamics, population variation, and clinical diagnostics.

Methods

Selection and prioritization of tools for TR benchmarking

In the initial phase, we reviewed the literature to identify software tools developed for TR genotyping. The primary source of TR methods was the long-read-tools.org resource [42,43], which was supplemented by searching Google Scholar for references to TR genotypers, including those in methods sections of analysis papers. We curated key characteristics, including supported sequencing technologies, handling of compound repeat structures and non-canonical motifs, detection of novel loci, and the nature of the output (haplotype-resolved genotypes, full allele sequences, and supported motif sizes), which yielded a list of 25 tools (**Additional file 1: Table S1**). For benchmarking, we selected TR genotypers that report full

allele sequences and are under active development (at least one update within the year preceding 11 November 2025). We also added the most popular length-only genotyper, straglr, for comparison. The final selection included 8 tools. (**Additional file 1: Table S2**).

Selection of the catalog for benchmarking

The TRExplorer catalog is a comprehensive TR catalog built from merging individual catalogs, including perfect TRs, adotto benchmark loci, polymorphic loci, VNTR catalogs, and disease-associated TR loci [37]. We used Version 2 of the TRExplorer catalog, which comprises 5.5 million loci in the human genome, of which ~100k are variation clusters: regions with closely spaced polymorphic TR loci. This catalog is large in terms of the time required for genotyping individual tools and also poses tool-specific problems during reformatting. For the first phase of benchmarking, we limited our analyses to a subset of loci that were non-overlapping to reduce genotyping complexity. Specifically, we selected non-overlapping loci separated by a minimum distance of 25bp to obtain a set of isolated TR loci. We then sampled 1000 loci for each motif size in the set. We ensured that disease-associated loci were included by adding the STRchive loci [5] back into our testing catalog. This gives us a total of 43,009 loci. The final benchmark catalog was designed to remain genome-wide despite subsetting, retaining loci across all chromosomes (**Additional file 2: Fig. S13**).

Sequencing data preparation

The ONT sequencing data for the seven GIAB samples (HG001-7) were downloaded from EPI2ME's Genome in a Bottle Data Release 2025.01. Each sample is sequenced on 2 flow cells, and the data is available as aligned Compressed Reference-oriented Alignment Map (CRAM) files. One CRAM file for one flow cell run of the sample, hence 2 CRAM files for a sample. For each sample, the data was pooled across both flow cells and downsampled to near 30x coverage. The CRAM files of both the flow cells for each sample are merged using `samtools merge` followed by `samtools view -s 42.{fraction}` for downsampling to 30x.

For the HPRC cohort, using the HPRC Data Explorer [69], we selected ONT runs expected to have ~30-35x sequencing depth in a single flow cell. We balanced samples sequenced on R9 and R10 chemistry: 33 R9.4.1 chemistry and 34 R10.4.1 for a total of 67 samples:

HG00290 HG01786 HG02630 HG04228 NA20129 HG00320 HG01975 HG02698 NA18505
NA20282 HG00321 HG01993 HG02723 NA18508 NA20346 HG00329 HG02004 HG02818
NA18608 NA20762 HG00350 HG02040 HG02922 NA18620 NA20806 HG00408 HG02071
HG02984 NA18948 NA20809 HG00597 HG02132 HG03017 NA18952 NA20870 HG00609
HG02135 HG03209 NA18960 NA21093 HG00639 HG02280 HG03453 NA18974 NA21106
HG01081 HG02293 HG03521 NA19240 NA21110 HG01255 HG02392 HG03710 NA19682
NA21309 HG01346 HG02572 HG03804 NA19776 HG01433 HG02583 HG03816 NA19835
HG01496 HG02602 HG04157 NA19909

ONT reads were extracted using `samtools` and aligned to GRCh38 using `minimap2` (see https://github.com/dashnowlab/TR-Benchmarking/blob/main/align_HPRC/main_align.nf for more details).

```
samtools fastq -T MM,ML ${unaligned_bam} | \
minimap2 -a -y -t ${threads} -R "${readgroup}" ${ref} - | \
samtools sort -@ ${threads} -O CRAM --reference ${ref} -T ${sample}.tmp -o $
{sample}.cram
```

Coverage estimation was done using mosdepth v0.3.10 [70] on each alignment file.

Implementation and execution of selected TR genotyping tools

All benchmarking workflows were implemented using the Nextflow workflow management system (DSL2), enabling scalable and reproducible execution across high-performance computing environments. Each Nextflow process was defined with a fully self-contained environment, avoiding reliance on inherited shell modules and preventing dependency conflicts to ensure consistent behavior across compute nodes. Tool dependencies were managed using either Singularity container images or Conda environments, depending on software compatibility: mosdepth, LongTR, Straglr, and STRkit were deployed via containers (Wave/Sequera community images or GitHub Container Registry), whereas ATaRVa, Medaka Tandem, and vamos required dedicated Conda environments; STRdust was executed as a statically pre-compiled binary. Several software bugs and compatibility issues were encountered during implementation and were reported to the respective tool developers. Eight TR genotypers (ATaRVa, LongTR, Medaka Tandem, Straglr, STRdust, STRkit, TRsv, and vamos) were evaluated using a catalog of 43,009 loci. We successfully generated results for seven tools. TRsv was excluded from downstream analyses due to persistent and reproducible runtime errors originating from its distributed Perl scripts and the Singularity container provided in the authors' repository, which prevented successful execution on our datasets. The final benchmarking pipeline was applied to 7 GIAB samples, 67 HPRC samples, and 31 samples from the Stevanovski 2022 cohort [36].

ATaRVa

ATaRVa (v0.5.0) was run using a minimum of one supporting read per allele (--min-reads 1) parameter and passed the sample karyotype.

LongTR

We evaluated LongTR (v1.2) using --alignment-params -1.0, -0.458675, -1.0, -0.458675, -0.00005800168, -1.0, -1.0, thereby increasing the match-to-insertion and match-to-deletion transition probabilities relative to the default settings to better accommodate the indel-rich error profile of long-read sequencing at TR loci as recommended by the authors. Prior to calling, we computed the maximum TR length in the locus catalog and passed it as the --max-tr-len parameter to ensure that LongTR could accommodate the longest repeat tract in our benchmark catalog. For samples with XY karyotype, chrX and chrY were treated as haploid by passing --haploid-chrs chrX,chrY.

Medaka Tandem

We used the Medaka (v2.1.1) tandem-repeat calling module: Medaka Tandem (tandem.py,

<https://github.com/nanoporetech/medaka/tree/master/medaka/tandem>). Sample sex was passed to Medaka as male or female based on the sample karyotype. For CRAM files that had been aligned in a different environment, Medaka runs failed with htslib cram_get_ref errors because the CRAM header UR tag pointed to an obsolete reference FASTA path, causing the underlying library to attempt to load a non-existent reference despite the correct GRCh38 FASTA being supplied on the command line. We had to re-header these samples to produce Medaka output.

<https://github.com/nanoporetech/medaka/issues/574>

Straglr

Straglr (v1.5.5) was executed using a Python wrapper, with alleles genotyped by size in base pairs (--genotype_in_size). Sample sex was passed as m or f based on sample karyotype.

STRdust

STRdust (v0.24.2) was run requesting unphased genotypes (--unphased). For samples with karyotype XY, we passed --haploid chrX,chrY so that STRdust treated chrX and chrY as haploid, while autosomes in all samples were modeled as diploid.

STRkit

STRkit (v0.24.2) was run in targeted calling mode with ploidy passed per sample (--ploidy) and sex chromosome karyotype supplied via --sex-chr. Local realignment around TR loci was enabled (--realign).

Vamos

vamos (v3.0.5) and (v2.1.7) were run using read-based calling (--read) with the addition of -S to include allele read consensus sequences in the VCF output. These consensus sequences reflect the raw output of the read consensus algorithm, prior to motif set interpretation. Sample allele sequences are reported in the VCF in a non-standard format, appended to the end of the sample field. The reference allele is always "N", and all GTs are 1/1 or 1/2; therefore, to use this VCF with downstream tools, we extracted the reference allele sequence from the reference genome using the locus start and end coordinates, compared sample alleles to the reference, and marked perfectly matching alleles as "0". Any non-reference alleles were moved to the ALT field. This post-processing was implemented in a custom fix-vcf.py script, which must be adjusted for the two versions of vamos separately to produce valid VCF format for downstream analysis.

To evaluate sensitivity to pathogenic repeat expansions, all seven tools were additionally run with minimum supporting-read thresholds. In this configuration, ATaRVa and LongTR required at least one supporting read (--min-reads 1), LongTR additionally applied a minimum mean base quality of 1 (--min-mean-qual 1), Medaka used a minimum depth of 1 (--min_depth 1), Straglr required at least one supporting read (--min_support 1) with a minimum cluster size of 1 (--min_cluster_size 1) and a maximum of two clusters per locus (--max_num_clusters 2), STRdust required at least one supporting read (--support 1), and STRkit required at least one supporting read per locus (--min-reads 1). All other parameters remained identical to the standard runs described above. Vamos was run with identical settings to the standard run, as it

does not expose a minimum read support parameter.

Extracting allele information from VCFs

Each tool uses slightly different INFO and FORMAT fields in the VCF. For tools that encode alleles as explicit nucleotide sequences in the REF and ALT fields (all except Straglr), allele sequences for each sample were resolved by mapping the GT indices to the corresponding REF or ALT entry, and loci with missing or unresolvable genotypes were excluded. For Straglr, which does not report explicit allele sequences but instead uses symbolic ALT entries, allele lengths were derived directly from INFO field annotations. The reference allele length was obtained from the RUL_REF INFO field, where available, with sequential fallback to $END - POS + 1$. ALT allele lengths were extracted from the RB INFO field, which records the repeat block length in base pairs for each alternative allele, and GT indices were used to assign the two allele lengths per sample.

Benchmarking against assembly

We used polished hybrid ONT/HiFi assemblies generated by the HPRC, which integrated 60x PacBio HiFi data with 30x ultra-long (>100 kbp) ONT data and phasing using Illumina trio-WGS or Hi-C [71]. To benchmark read-based genotyping against assemblies, we downloaded ~30-35x ONT reads from HPRC for 67 samples: 33 samples with data generated using the R9.4.1 chemistry and 34 using the R10.4.1 chemistry, along with their corresponding hybrid assemblies. Phased HPRC assemblies were aligned to the GRCh38 reference genome using minimap2 with the `-x asm5` preset to generate BAM files. Allele lengths at each locus were calculated by lifting over locus coordinates from the reference to the sample assemblies using these alignments, implemented in an in-house Python script (`assembly_concordance.py`; https://github.com/dashnowlab/TR-Benchmarking/scripts/assembly_concordance.py).

Some tools adjust the coordinates for a subset of input TR loci. STRkit scans the flanking sequences of each TR locus in the reference for possible repeat extensions and extends the coordinates if matching motifs are found. LongTR expands locus coordinates to include flanking sequences in which sequence variations are identified. For assembly-based concordance, genotypes are therefore evaluated using tool-specific coordinates by performing liftover on the coordinates as reported in each tool's output VCF. Only loci with single-contig alignments between the reference and the haploid assemblies were retained, and diploid genotypes were constructed by calculating allele lengths independently from each haplotype.

The ONT read data for each sample is aligned to GRCh38 reference genome using minimap2 [72] and the catalog loci are genotyped using all seven tools. Allele sequences or allele lengths (in the case of Straglr, which does not report sequences) were extracted from each tool's VCF (See Extracting allele information from VCFs), and the corresponding assembly alleles were extracted based on the VCF coordinates. For each genotype, alleles called by each tool were paired with assembly alleles such that the combined deviation or error was minimized. For tools reporting allele sequences, deviation was calculated as Levenshtein distance; a measure of the

minimum number of single-nucleotide edits (insertions, deletions, or substitutions) required to transform one sequence into another. For Straglr, deviation was calculated as the absolute difference in allele length. Length deviations were categorized into four concordance classes: Match (no deviation), Off by 1 bp (deviation of exactly 1 bp), Off by motif (deviation $\leq$ the motif length of the locus), and Mismatch (deviation exceeding the motif length) (**Additional file 2: Fig. S1**). For each sample and tool, this procedure generates a per-locus concordance report containing the tool-called and assembly-called alleles, the length deviation, the Levenshtein distance, and the concordance class. This report serves as the basis for all subsequent assembly concordance analyses.

Mendelian consistency

To assess the accuracy of TR genotyping, we evaluated Mendelian consistency across the parent–offspring HG002 trio. HG002 (the offspring) is male (46, XY); because loci on the X and Y chromosomes are hemizygous in the offspring, sex chromosomes were excluded, and Mendelian consistency was assessed exclusively on autosomes. Alleles for each sample in the trio are extracted (See Extracting allele information from VCFs). Loci where any trio member had a missing call were excluded. Allele lengths in base pairs were then derived from the resolved sequences, and each individual's genotype was encoded as a four-dimensional feature vector $[L_1, L_2, \Delta L_1, \Delta L_2]$, where L_1 and L_2 are the lengths of the two alleles and ΔL_1 and ΔL_2 are their deviations from the reference allele length ($L_{ref} = \text{len}(\text{REF})$). For Straglr, the same four-dimensional feature vector was constructed based on the two allele lengths.

In both scripts, VCFs for the trio members were stream-merged by chromosomal position (CHROM, POS), with LongTR merged by variant ID due to its coordinate representation. Loci present in any subset of trio members were processed only when both parental records were available. Mendelian consistency was assessed by enumerating all four possible parental transmission combinations (one allele from each parent) and both orderings of the offspring's allele pair, yielding a total of eight candidate pairings. For each pairing, the Manhattan distance between the parental transmission feature vector and the offspring feature vector was computed, and the minimum across all pairings was taken as the locus-level inheritance distance. A locus was classified as Mendelian-consistent (MATCH) if this minimum distance equalled zero, indicating that the offspring's two allele lengths were exactly explained by one allele from each parent. To capture near-consistent genotypes, we additionally defined three lenient categories: ONE-OFF, where the minimum distance corresponded to a single base pair discrepancy per allele (half-distance = 1 bp); MLEN-OFF, where the discrepancy was within one motif length unit; and MISMATCH for all remaining loci. Analyses were stratified by offspring genotype class (0/0, 0/1, 1/1, and 1/2) and by repeat motif length, with the latter binned into unit categories of 1–10 bp and a ">10 bp" group. Mendelian consistency rates were reported both per genotype class and aggregated across loci, with the combined rate of MATCH, ONE-OFF, and MLEN-OFF referred to as the "good" rate.

Concordance between genotypers

VCF output files from each tool are compared using the `run_comparisons.py` script available at <https://github.com/dashnowlab/TR-Benchmarking>. Some tools change the start and/or end coordinates of loci from those provided as input. To ensure comparisons occur at the same genomic positions across files, the catalog is used as a positional reference. Before comparisons are performed for a given line, each tool locus is assigned to the overlapping catalog position. If a VCF does not contain a matching locus, that tool is skipped until its coordinates align with the catalog again.

Once synchronized, the script first determines if the tools are widening or narrowing the coordinate boundaries relative to the catalog. Finally, the specific alleles from each tool are compared using length differences and the Levenshtein distance (a metric representing the total insertions, deletions, and substitutions needed to transform one allele into the other). Straglr did not provide sufficient information to reconstruct a sequence for Levenshtein distance analysis. Consequently, only length and positional comparisons were performed for that tool.

Pathogenic loci

We analyzed previously published Stevanovski 2022 cohort [36] blood-derived ONT sequencing data from 27 individuals with a diagnosis and/or allele-size estimates from molecular testing using RP-PCR, Southern blot, or both (**Additional file 1: Table S8**). Targeted ONT sequencing was generated using adaptive sampling of 37 target genes with known TR expansions (see “Supplementary Table 1” in [36]), including those with expansions in the samples: *ATXN1*, *C9orf72*, *DMPK*, *FMR1*, *FXN*, *HTT*, *NOTCH2NLC*, *PABPN1*, and *RFC1*.

Raw reads were aligned to GRCh38 using minimap2 and genotyped with all seven tools using the same pipeline and ~43k TR catalog as described above. We ran two versions of *vamos*, v2.1.7 and v3.0.5, because a new version was released during the analysis phase of this study. We then subset the results to only the relevant disease locus for each sample by filtering to VCF rows with a starting position within +/- 50 bp of the disease locus start position in STRchive-disease-loci-v2.15.0.hg38.general.bed (<https://github.com/dashnowlab/STRchive/releases/tag/v2.15.0>) [5].

The *PABPN1* locus has different pathogenic thresholds in the literature: 8 vs 12, depending on whether you include all GCN polyalanine codons or only pure GCG repeats [73]. The RP-PCR validations in [36] used the narrower definition, while STRchive [74] uses the wider definition; therefore, we added 4 motifs to the RP-PCR values to make them consistent with the modern definition.

Tool sensitivity to pathogenic or premutation expansions was assessed by comparing the length of the allele in the tool VCF with the pathogenic threshold (pathogenic min) or, for premutation alleles, the intermediate threshold (intermediate min) from STRchive. If the tool allele length was greater than or equal to the threshold, it was marked as a successful detection. For dominant diseases, one allele above the threshold was required; for recessive diseases, two. For X-linked conditions, if tools reported two alleles, at least one allele above the threshold was required in

hemizygous males. Some tools do not accept sample sex as input.

Availability of data and materials

All sequencing datasets analyzed in this study are publicly available and were obtained from reference genomes generated by the Genome in a Bottle (GIAB) consortium. Long-read whole-genome sequencing data for samples HG001–HG007 were obtained from Jan 2025 sequencing data released by EPI2ME of GIAB samples (s3://ont-open-data/giab_2025.01/). HPRC genomes were identified using the Data Explorer (<https://data.humanpangenome.org/raw-sequencing-data>) and downloaded from their AWS s3 bucket (s3://human-pangenomics/). Adaptive sampling targeted sequencing of samples with pathogenic expansions were downloaded from the NCBI Sequence Read Archive (SRA) under accession no. PRJNA786382. All other data supporting the findings of this study are available either within the article, the supplementary information, or the supplementary data files.

The downsampled CRAMs generated in this study have been deposited in the European Nucleotide Archive (ENA) at EMBL-EBI under accession number PRJEB114757 ([ebi.ac.uk](https://www.ebi.ac.uk)) [75] and the VCFs have been deposited in Zenodo with the DOI: 10.5281/zenodo.20332762 [76]. For convenience, these files are also available on AWS at https://s3.amazonaws.com/1000g-ont/index.html?prefix=working_dir/tr_benchmarking/.

The study utilized previously published analysis tools and custom scripts described in the Methods section. Custom scripts developed for this study are available in a public repository at <https://github.com/dashnowlab/TR-Benchmarking> and have been archived (version 1.0.1) on Zenodo with the DOI: 10.5281/zenodo.20292923 [65]. The custom scripts and workflow components provided in this study are intended as templates to support reproducibility and reuse. Users are expected to modify file paths, software environments, and other run-specific parameters as needed for their local system and input data.

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Contributions

Contributions are described following the <https://credit.niso.org/> framework.

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All authors read and approved the final manuscript.

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Fritz Sedlazeck and Claudia Feng were the primary editors of this article and managed its editorial process and peer review in collaboration with the rest of the editorial team. The peer-review history is available in the online version of this article.

Ethics declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

DEM is on scientific advisory boards at Basis Genetics and Inso Biosciences, is engaged in research agreements with Oxford Nanopore Technologies (ONT), PacBio, and Illumina, and has received research and travel support from ONT, PacBio, and Illumina. DEM receives research funding from BioMarin for serving as the site PI on a clinical trial. DEM holds stock options in MyOme, Basis Genetics, and Inso Biosciences, and is a consultant for MyOme.

The remaining authors declare no competing interests.

Supplementary information

Additional file 1: Supplementary Tables 1-8

Additional file 2: Supplementary Notes and Supplementary Figures 1-13

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